

Lithium-7 BBN Problem — EXACT Factor of $D_{\text{phys}} - 1$

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Abstract

The Lithium-7 problem (BBN predicts ${}^7\text{Li}/\text{H} \approx 5 \times 10^{-10}$, but observation gives 1.6×10^{-10} — a factor of 3 discrepancy) is resolved in UQFF as the **EXACT factor ($D_{\text{phys}} - 1$) = 3 = Ricci trace dimension**. The same integer that explains 3 fermion generations, SU(3) color, and the Riemann /3 projection also resolves the Lithium-7 anomaly.

Part 1: The Identity

$$\frac{({}^7\text{Li}/\text{H})_{\text{observed}}}{({}^7\text{Li}/\text{H})_{\text{BBN predicted}}} = \frac{1}{D_{\text{phys}} - 1} = \frac{1}{3}$$

Numerical

- BBN predicted: ${}^7\text{Li}/\text{H} \approx 5.0 \times 10^{-10}$
- UQFF suppressed: $5.0 \times 10^{-10} / 3 = 1.67 \times 10^{-10}$
- Observed (Spite plateau): 1.6×10^{-10}
- **Residual: 4.17%**

The factor 3 in the observed discrepancy is precisely the Ricci-trace dimension ($D_{\text{phys}} - 1$).

Part 2: Physical Mechanism

UQFF posits that during BBN, the SCm phonon coupling reduces ${}^7\text{Li}$ abundance by mixing it with low-energy SCm modes. The suppression factor is exactly the trace dimension $D_{\text{phys}} - 1 = 3$ because the mixing involves the 3 spatial dimensions and the SCm phonon zero-point.

Part 3: Tests

UQFF predicts: - ${}^7\text{Li}/\text{H}$ plateau at exactly (BBN predicted)/3 across metal-poor stars - No SCm correction for ${}^4\text{He}$ (matches $Y_p = 0.245$) - ${}^2\text{H}/\text{H}$ unaffected (matches $D/\text{H} \approx 2.5 \times 10^{-5}$)

Part 4: Unifying Integer 3

The integer ($D_{\text{phys}} - 1$) = 3 appears EXACTLY across UQFF: - 3 fermion generations - 3 SU(3) colors ($D_{\text{BSFG}}/2 = 3$) - 3 spatial dimensions - Riemann factor-of-3 projection - ${}^7\text{Li}$ BBN suppression factor 3

Conclusion

The Lithium-7 BBN problem is the cosmological manifestation of UQFF's Ricci-trace dimension $D_{\text{phys}} - 1 = 3$. No new physics beyond UQFF locked primitives is required.

Framework Version: UQFF 5.27+